

Drawing with mckPy

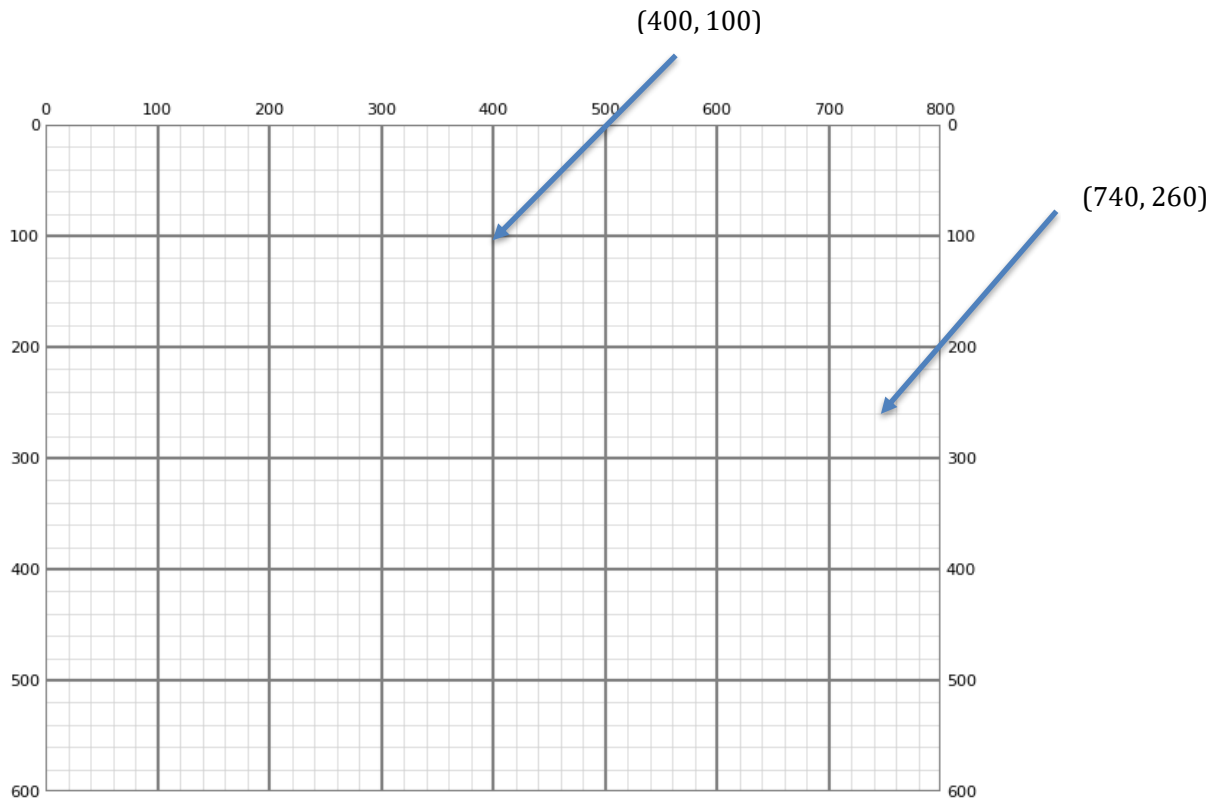
Basic Setup

All programs that use mckPy.py have the same basic setup. At minimal you need to import mckPy.py and have a while running(): loop. mckPy.py must be in the same folder as your program.

```
from mckPy import *  
  
while running():  
    drawRect("red", 350, 280, 100, 60)
```

Shape Drawing Functions

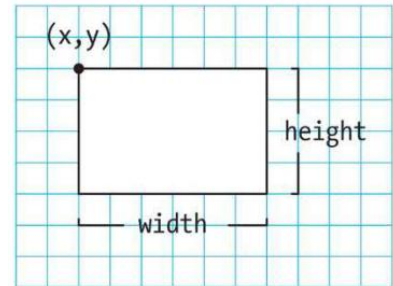
The screen is made up of pixels. Each pixel has an (x,y) location. By default, the screen is 800 x 600.



drawRect(colour, x, y, width, height)

Draws a rectangle with the top-left corner at (x, y), and dimensions of width x height.

e.g., `drawRect("red", 200, 100, 50, 40)`



options

thickness

`drawRect("red", 200, 100, 50, 40, 5)`

- will draw an unfilled box with a 5 pixel outline



outline

`drawRect("skyblue", 200, 100, 50, 40, outline = 3)`

- will draw a filled box with a 3 pixel outline



corner

`drawRect("skyblue", 200, 100, 50, 40, corner = 15)`

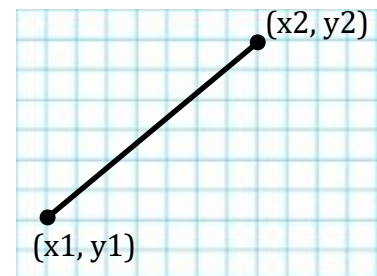
- will draw a filled box with rounded corners. The corner starts rounding 15 pixels from the corner



drawLine(colour, x1, y1, x2, y2)

Draws a line from the point (x1, y1) to (x2, y2).

e.g., `drawLine("black", 10, 70, 80, 10)`



options

thickness

`drawLine("red", 10, 70, 80, 10, 5)`

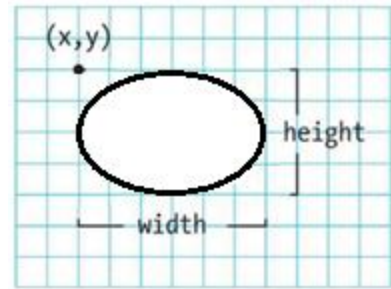
- Makes the line 5-pixels thick.



drawOval(colour, x, y, width, height)

Draws an oval inside the area specified by the rectangle with the top-left corner at (x, y), and dimensions of width x height.

e.g., `drawOval("black", 200, 100, 50, 40)`



options

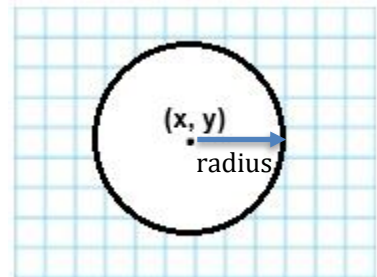
thickness, outline

These work the same way they do for rectangle.

drawCircle(colour, x, y, radius)

Draws a circle, centered at (x, y) with the given radius.

e.g., `drawCircle("black", 200, 100, 50)`



options

thickness, outline

These work the same way they do for rectangle.

drawTriangle(colour, x1, y1, x2, y2, x3, y3)

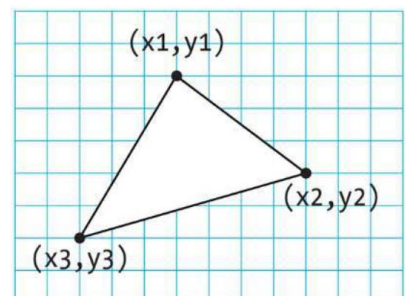
Draws a triangle that connects the points (x1, y1) (x2, y2) (x3, y3) .

e.g., `drawTriangle("blue", 50, 20, 90, 50, 20, 70)`

options

thickness, outline

These work the same way they do for rectangle.

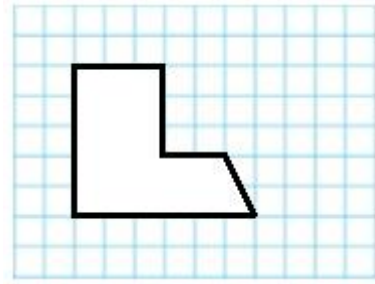


drawPoly(colour, points)

Draws a polygon with as that connects the points given. These points must trace the outline in a continuous order.

e.g.,

```
points = [(20, 20), (50, 20), (50, 50), (70, 50), (80, 70), (20, 70)]  
drawPoly("green", points)
```



options

thickness, outline

These work the same way they do for rectangle.

e.g.,

```
drawPoly("green", points, outline = 3)
```

fill(colour)

Fills the whole screen in the given colour.

e.g., `fill("beige")`

resize(width, height)

Changes the size of the screen. Make sure you place this BEFORE your while running().

e.g., `resize(800, 800)`

Text Functions

drawText(colour, text, x, y)

Draws text on the screen in the specified colour, in the current font. The default font is "Helvetica", 24. The x,y will be the top-left corner of the text.

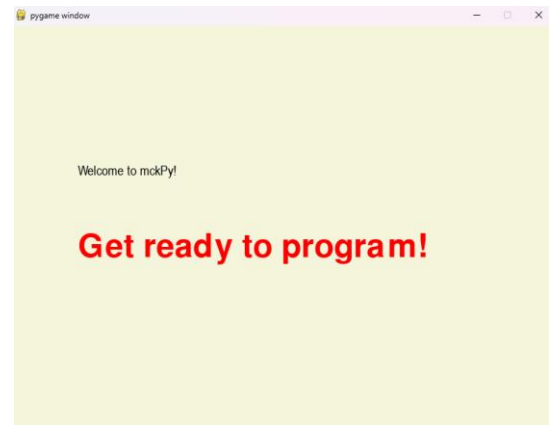
setFont(font_name, font_size)

Changes the current font.

e.g.,

```
from mckPy import *

while running():
    fill("beige")
    setFont("Arial", 20)
    drawText("black", "Welcome to mckPy!", 100, 200)
    setFont("Britannic Bold", 70)
    drawText("red", "Get ready to program!", 100, 300)
```



getText(x, y, width , colour = "skyblue")

Draws an input box on the screen and lets the user type in text, using enter to end the input. The height of the box is based on the current Font.

```
if clickOnRect(320, 400, 160, 80):
    name = getText(200, 100, 400)

drawText("white", name, 350, 10)
```

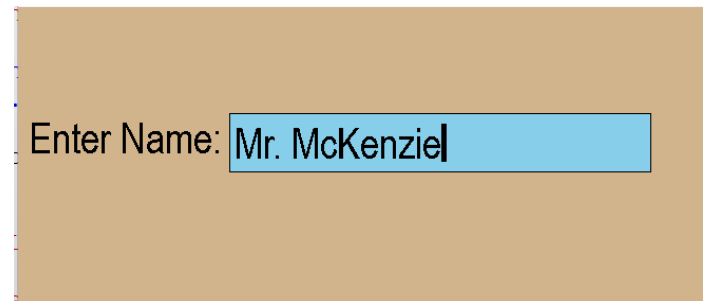


Image Functions

loadImage(image_name)

Loads an image into a variable to be drawn later with the drawImage command.

e.g.,

```
backImg = loadImage("back.jpg")
```

options

width, height

```
backImg = loadImage("back.jpg", 800, 600)
```

- This will scale the image to size 800 x 600.

drawImage(image, x, y)

Draws an image that was loaded with loadImage.

e.g.,

```
from mckPy import *

background = loadImage("space.jpg", 800, 600)
ship = loadImage("ship.png")

while running():
    drawImage(background, 0, 0)
    drawImage(ship, 200, 300)
```

